Diagnosing Quadrant and Parameter-Orientation Errors Answer Key

Precalculus — Polar and Parametric Representations

Quick Answers

1. 126.87	3. -3.46	5. 30, 150	7. 20
2. 315	4. 2.60	6. 247.38	8. —

Curriculum

Level: Precalculus — Polar and Parametric Representations
HSF-TF.A.2, HSF-TF.B.5 – *problems 1, 2, 3, 4, 6, 7, 8*
HSF-TF.C.8 – *problem 5*
Difficulty 2–5 of 5 across 8 tagged checks.

Common wrong answers

1. *If they got -53.13: reported the calculator's arctan(y/x) without adjusting for Quadrant II*
2. *If they got 135: added 180 to the calculator's -45 instead of 360, landing in Quadrant II*
3. *If they got 3.46: treated r = -4 as 4, ignoring the reflection through the pole*
4. *If they got 1.5: read the parametrization backwards and used x = 3cos t*
6. *If they got 67.38: reported the reference angle and never placed the point in Quadrant III*
7. *If they got 380: added 180 instead of subtracting, leaving the angle outside [0, 360)*

Each solution names the diagnosis first, then the repair. Give credit for the diagnosis separately from the number: a student who fixes the value but cannot say why the first answer failed will repeat the error. Angles are rounded to two decimal places.

1. $P(-3,4)$: Devin reports $(5,-53.13°)$.
Diagnosis: $r = 5$ is right. The angle is not: arctan returns values in $(-90°, 90°)$, i.e. Quadrant I or IV only, so it cannot report a Quadrant II angle. $P(-3,4)$ has $x < 0$ and $y > 0$, so P is in Quadrant II — and $-53.13°$ points into Quadrant IV, the exact opposite direction.
Repair: keep the reference angle $\arctan\left(\frac{4}{3}\right) = 53.13°$ and place it in Quadrant II: $\theta = 180° - 53.13°$. $\boxed{\theta = 126.87°}$

So $P = (5,\ 126.87°)$.

2. $Q(2, -2)$: the classmate reports $135°$.
Diagnosis: $x > 0$ and $y < 0$ put Q in Quadrant IV, but $135°$ is in Quadrant II. The calculator's $-45°$ was actually correct in direction — it just was not in $[0°, 360°)$. Adding $180°$ reverses a direction; adding $360°$ renames it.
Repair: $\theta = -45° + 360°$, i.e. $360° - 45°$. $\boxed{\theta = 315°}$

With $r = \sqrt{2^2 + (-2)^2} = 2\sqrt{2} \approx 2.83$, $Q = (2\sqrt{2},\ 315°)$.

3. $(-4, 60°)$ to rectangular: Sasha reports $(2, 3.46)$.
Diagnosis: she used $r = 4$. A negative r means "go the given direction, then straight through the pole": the point $(-4, 60°)$ is the same as $(4, 240°)$, which lies in Quadrant III — both coordinates negative. Her answer sits in Quadrant I.
Repair: substitute the signed r into $y = r \sin\theta$: $y = -4 \sin 60° = -4(0.8660\ldots)$.
$\boxed{y = -3.46}$

($x = -4\cos 60° = -2$, so the point is $(-2,\ -3.46)$.)

4. $x = 3\sin t$, $y = 3\cos t$ at $t = \frac{\pi}{3}$: the student reports $x = 1.5$.
Diagnosis: $1.5 = 3\cos\frac{\pi}{3}$ — the student read the parametrization backwards and used the y-rule for x.
Repair: $x = 3\sin\frac{\pi}{3} = 3\left(\frac{\sqrt{3}}{2}\right) = 2.598\ldots$ $\boxed{x = 2.60}$
Orientation: $x = 3\cos t$, $y = 3\sin t$ starts at $(3, 0)$ and runs *counterclockwise*; $x = 3\sin t$, $y = 3\cos t$ starts at $(0, 3)$ and runs *clockwise*. Same circle, different starting point and opposite direction — so the two parametrizations are not interchangeable.

5. $x = 4\cos t$, $y = 2\sin t$ on $0° \le t < 360°$: the student answers $t = 30°$ only.
Diagnosis: $y = 1$ gives $\sin t = \frac{1}{2}$, and arcsin returns only the Quadrant I solution. The curve is traced all the way around, so it reaches height $y = 1$ once on the way up the right side and once on the way down the left side. One solution can only be the whole answer if the parameter interval is restricted.
Repair: $2\sin t - 1 = 0 \Rightarrow \sin t = \frac{1}{2}$, so $t = 30°$ and the supplementary value $t = 180° - 30°$.
$\boxed{t = 30° \text{ and } 150°}$
(Check the points: $t = 30°$ gives $(3.46, 1)$ and $t = 150°$ gives $(-3.46, 1)$ — two different points at the same height, as the ellipse requires.)

6. $R(-5, -12)$: the student reports $(13, 67.38°)$.
Diagnosis: both coordinates are negative, so R is in Quadrant III. The two minus signs cancelled inside the fraction, $\frac{-12}{-5} = \frac{12}{5}$, so the calculator returned the Quadrant I reference angle. arctan cannot distinguish Quadrant I from Quadrant III — the signs of x and y must be read off separately.
Repair: reference angle $\arctan\left(\frac{12}{5}\right) = 67.38°$, placed in Quadrant III: $\theta = 180° + 67.38°$.
$\boxed{\theta = 247.38°}$

So $R = (13,\ 247.38°)$.

7. $(-6, 200°)$ with positive r: the student writes $(6, 380°)$.
Diagnosis: flipping the sign of r does require a half-turn, but a half-turn can be taken in either direction. Adding $180°$ to $200°$ leaves $380°$, which is outside the required interval $[0°, 360°)$; subtracting $180°$ lands the same direction inside it. (Indeed $380°$ and $20°$ name the same ray, so the direction was right and only the naming was wrong — but an answer outside the stated interval is not yet an answer.)
Repair: $\theta = 200° - 180°$. $\boxed{\theta = 20°}$
So $(-6,\ 200°) = (6,\ 20°)$.

8. $C_1 : x = 2 + 3\cos t,\ y = -1 + 3\sin t$ and $C_2 : x = 2 + 3\cos(-t),\ y = -1 + 3\sin(-t)$, $0 \le t < 2\pi$.
Model answer (open reasoning — review by hand):
Same curve. Since $\cos(-t) = \cos t$ and $\sin(-t) = -\sin t$, C_2 is $x = 2+3\cos t$, $y = -1-3\sin t$. For both curves $(x-2)^2 + (y+1)^2 = 9\cos^2 t + 9\sin^2 t = 9$, so both trace the circle of radius 3 centred at $(2, -1)$, and both cover it completely as t runs over one period.
Opposite orientation. Build the table:

t	C_1	C_2
0	$(5, -1)$	$(5, -1)$
$\pi/2$	$(2, 2)$	$(2, -4)$
π	$(-1, -1)$	$(-1, -1)$
$3\pi/2$	$(2, -4)$	$(2, 2)$

Both start at $(5, -1)$, but after a quarter turn C_1 is at the top of the circle and C_2 is at the bottom: C_1 is counterclockwise, C_2 clockwise. On the sketch the arrows point opposite ways around the same circle.
Why the Cartesian equation cannot decide it. Eliminating t gives $(x-2)^2 + (y+1)^2 = 9$ for both. That equation is a *set* of points; it records no order, no starting point, and no direction. Orientation lives in the parameter, so only the parametric form — a table of increasing t, or the sign of dy/dt at a chosen point — can show it.
Accept any answer that (a) shows both curves satisfy the same circle equation, (b) exhibits at least two matched t values proving the directions differ, and (c) states that orientation is information the Cartesian equation does not carry.